Injecting Linguistic Into Visual Backbone: Query-Aware Multimodal Fusion Network for Remote Sensing Visual Grounding

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***Abstract*— The remote sensing visual grounding (RSVG) task focuses on accurately identifying and localizing specific targets in remote sensing (RS) images using descriptive query expressions. Existing methods independently extract visual and textual features, ignoring early complementary information between image and text. This leads to information loss and misalignment, limiting the model's ability to distinguish similar targets. To address this challenge, we propose the query-aware multimodal fusion network (QAMFN), which introduces an innovative query-guided visual attention (QGVA) mechanism in the early stages of the visual encoder. This mechanism integrates textual information during the early visual feature extraction process, thereby resolving the issue of missing image–text complementary information. QGVA ensures that the visual backbone accurately focuses on local features highly relevant to the query by injecting textual information into the visual encoding process. Additionally, to enhance the model's ability to integrate multimodal information and adapt to more complex RS images, we introduce the text-semantic attention-guided masking (TAM) module. TAM aggregates multimodal features processed by the backbones and filters out redundant information, producing high-quality fused features. Experiments demonstrate that our approach sets a new record on the DIOR-RSVG dataset, improving accuracy to 81.67% (an absolute increase of 4.98%).**

***Index Terms*— Multimodal learning, query-aware multimodal fusion network (QAMFN), remote sensing (RS) visual grounding.**

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I. Introduction

In RECENT years, the quick development of remote sensing (RS) science has significantly enhanced our ability to perceive the Earth [1], [2], [3]. To effectively extract information from the vast amount of RS data, researchers have combined natural language with RS images, undertaking tasks such as RS image caption [4], [5], [6], RS cross-modal retrieval [7], [8], [9], and RS visual question answering [10], [11], [12]. However, extracting target information from single RS images with numerous targets and complex backgrounds remains challenging [13], [14]. Consequently, the exploration of visual grounding tasks in RS scenes has commenced. Visual grounding in RS images (RSVG) aims to identify object locations through textual queries, marking indicated objects with bounding boxes. It is crucial for research and has broad applications in emergency coordination, traffic, infrastructure, farming surveillance, city design, and agriculture management [5], [10], [15].

Several studies pertaining to RSVG have been introduced. Sun et al. [14] pioneered the concept and developed GeoVG. GeoVG is composed of three main components: a linguistic encoder, an image encoder, and a fusion module. The linguistic encoder interprets numerical geospatial relationships and converts complex expressions into geospatial relation graphs. The image encoder focuses on understanding large-scale RS scenes using adaptive region attention. Finally, the fusion module integrates text and image features for effective visual grounding. Zhan et al. [13] introduced a transformer-based MGVLF for RSVG, utilizing multiscale visual characteristics and text embeddings at various granularities to enhance RSVG precision.

Although the current RSVG methods have achieved some success, they all overlook early-stage complementary information between text and images, resulting in insufficient alignment of textual and visual features, thereby limiting the localization performance of the models. In Fig. 2, some examples of mislocalization are presented: (a) for the query "the baseball field in the middle," the model located the baseball field in the upper left, appearing to ignore the term "middle"; (b) for the query "the large chimney," the model located the smaller chimney on the left, appearing to disregard the adjec-

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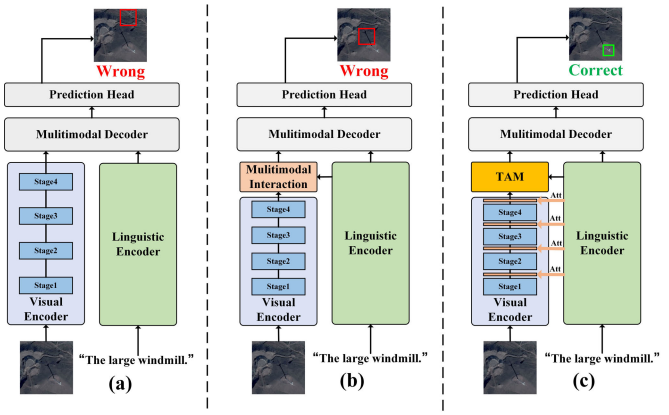


Fig. 1. Comparison between our proposed RSVG framework and previous RSVG frameworks. (a) Earliest RSVG framework encodes RS image and text, then uses a multimodal decoder to merge features and predict the final bounding box. (b) Current SOTA RSVG framework, which builds on (a) by incorporating multimodal interaction. (c) Our proposed RSVG framework, which integrates textual information during the visual encoding process, effectively utilizing the complementary information of image and text to achieve more accurate RSVG predictions.

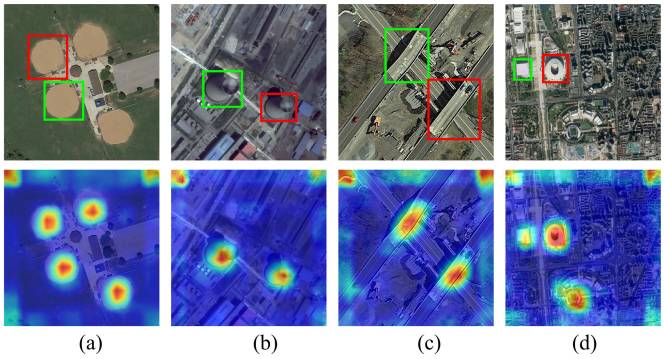


Fig. 2. Experimental results from previous RSVG method and feature map output by the visual backbone. The image above shows the inference results of the model, with the green box containing the ground truth and the red box containing the results inferred by the model. The image below presents the feature maps output by the pretrained visual backbone. (a) Baseball field in the middle. (b) Large chimney. (c) Overpass without moving vehicles. (d) Stadium to the left of the white road.

tive "large"; (c) for the query "the overpass without moving vehicles," the model located the overpass at the bottom, which has vehicles; and (d) for the query "the stadium to the left of the white road," the model located the stadium on the right instead. From these examples, it is evident that current RSVG methods, during visual grounding, seem to overlook certain features contained in the textual information, such as "to the left of the white road," "large," "middle," and "without moving vehicles." However, to accurately identify a specific feature in RS images, it is essential to consider not only the inherent characteristics of the feature itself but also its relationship with other features, as well as its positional and scale information within the RS scene. These detailed pieces of information are derived from the query text, and we refer to these crucial details as image–text complementary information. Image–text complementary information provides higher level features for the target feature, ensuring its uniqueness and preventing mislocalization.

As illustrated in Fig. 1(a) and (b), the existing RSVG methods use separate text and visual encoders to obtain textual and image features, respectively, and then perform prediction through cross-modal interaction to achieve feature fusion. This independent encoding neglects the early complementary information between text and image, resulting in incomplete utilization of detailed information in textual descriptions during feature extraction. For instance, when textual descriptions include positional information (e.g., "the baseball field in the middle") or size descriptions (e.g., "the large chimney"), models with independent encoding often struggle to accurately align these details during the later fusion process, resulting in mislocalization. Additionally, as seen from the feature maps in Fig. 2, many irrelevant targets are activated in the visual features generated by the visual encoder, which are unrelated to the textual information. This also introduces interference in the subsequent cross-modal interaction. Therefore, effectively utilizing early image–text complementary information to achieve more accurate RSVG remains a problem that needs to be addressed.

To address the aforementioned issues, we reexamined the process of extracting visual features. Visual features are typically extracted using convolutional neural networks (CNNs). As studied in [16], it has been shown that the shallow layers of a CNN have smaller receptive fields, allowing them to capture local detailed information. As the network deepens, the receptive field of the convolutional layers increases, thus capturing more global information. Local features can capture details and local information within an image, while global features can capture the overall context and global information of the image. In RSVG task, the detailed information contained in the query expressions, such as positional and scale information, can be better preserved in local features but is difficult to capture in global features. Therefore, to fully utilize image–text complementary information and achieve finer-grained feature alignment during feature extraction, the local features relevant to the query should be enhanced from the early stages of visual feature extraction to ensure their effective representation in the global features. Introducing textual information guidance in the early stages of visual feature extraction is an effective solution. By integrating textual information into the shallow convolutional layers, early interaction between image and text can be achieved, allowing the key information in the query text (such as location, size, and descriptive terms) to directly influence the local feature extraction process. For example, for the query expression "the stadium to the left of the white road," incorporating the positional information "to the left of the white road" during the shallow convolution process enables the network to focus more on the areas of the image related to "the left of the white road" during feature extraction. This enhances the feature representation of that specific area.

With this objective in mind, we propose a novel RSVG framework called the query-aware multimodal fusion network (QAMFN). As illustrated in Fig. 1(c), we have made improvements based on Fig. 1(b) by incorporating query attention guidance in the early stages of visual encoding to fully utilize early image–text complementary information and enhance the representation of visual information relevant to the

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query. Specifically, in QAMFN, we introduce the query-guided visual attention (QGVA) mechanism. QGVA leverages query information to enhance the visual feature map and integrates multiscale visual features under the guidance of textual features. We apply QGVA at various stages of the visual encoder to help focus on important regions within multiscale visual features. In addition, RS images contain many targets and cluttered backgrounds, and their visual features contain a lot of interfering information, which is not conducive to alignment with textual features. In order to align image and textual information more accurately, we need a more powerful multimodal interaction module. To this end, we design the text-semantic attention-guided masking (TAM) module, which uses a dynamic fusion method to effectively aggregate the global features of both visual and textual modalities, reduce irrelevant information in the visual features, and enhance the multimodal reasoning ability of the multimodal interaction part to better align the image and textual information.

In summary, the contributions of this article can be outlined as follows.

1) We propose a new RSVG framework called QAMFN. QAMFN introduces textual information guidance at various stages of visual encoding to customize high-quality visual features enriched with image–text complementary information. Our framework has been validated through experiments, achieving an accuracy of 81.96% on the Dior-RSVG dataset and establishing a new SOTA performance.
2) In QAMFN, we introduce the QGVA mechanism. QGVA operates during the visual encoding process, aligning with textual information in the early stages of visual feature extraction to customize high-quality visual features highly relevant to the query text.
3) To enhance the model's ability to integrate multimodal information and adapt to more complex RS images, we propose the TAM module. TAM enhances the model's multimodal reasoning ability and suppresses the expression of irrelevant information by using a dynamic fusion method to integrate the global features of both visual and textual modalities.

The remainder of this article is structured in the following manner. Section II presents the related work on RSVG. Section III elaborates in detail on our proposed QAMFN. Section IV presents comprehensive experiments and analyses. Finally, Section V wraps up this study and explores potential avenues for future research.

II. Related Works

In this section, we provide an extensive overview of the related work on visual grounding in natural images and RS images. More specifically, Section II-A delivers an intricate overview of methods in visual grounding for natural images. Section II-B summarizes the existing RSVG methods.

A. Visual Grounding Methods in Natural Images

1) Two-Stage Methods: Traditional two-stage visual grounding methods mainly consist of two phases: initially generating region proposals, followed by cross-modal matching. Typically, region proposals are generated by pretrained object detectors [17], [18]. Most traditional two-stage methods are exploring more effective ways of cross-modal interaction [19], [20], [21]. Yu et al. [20] decompose queries and images into three distinct modular components, pertaining to subject, location, and relation, and integrate cross-modal information within these three submodules using attention mechanisms. NMTree [22] and RvG-Tree [23] employ tree networks to parse language, a method that helps understand relationships between different parts of a sentence. Yang et al. [24] and Wang et al. [25] integrated graph attention networks for better modeling of cross-modal alignment. While two-stage methods in visual grounding have been notably successful, their efficacy largely depends on the quality of initial candidate boxes. These methods often involve sifting through numerous ineffective candidate boxes, leading to increased computational overhead and diminished efficiency.

2) One-Stage Methods: One-stage methods extract visual features while preserving spatial arrangement and conduct cross-modal interactions at the pixel level. One-stage methods exhibit higher computational efficiency and circumvent the error accumulation often encountered in multistage frameworks, thereby garnering significant research attention. For instance, Chen et al. [26] developed a multimodal interactant to explore the intricate associations between visual and textual characteristics. Yang et al. [27] encoded linguistic expressions into text embeddings and integrated them into the YOLOv3 detector [28], broadcasting the query embeddings to every spatial location. This approach produces dense object detections accompanied by confidence ratings, prioritizing the highest ranked objects for reference prediction. Liao et al. [29] conceptualized visual grounding as a correlation filtering process. In their approach, they aligned textual features with visual ones and subsequently applied correlation filtering to the visual feature. Yang et al. [30] proposed the ReSC framework to overcome the challenges posed by FAOA in intricate query visual grounding through multiple rounds of fusion. Liao et al. [31] introduced the PLV mechanism, further enhancing the inferencing capabilities of one-stage visual grounding approaches. Despite significant achievements, one-stage approaches usually employ point features for representing objects, which limits their performance in capturing specific and detailed object features.

3) Transformer-Based Methods: In recent years, transformer-based visual grounding methods have demonstrated their efficiency and simplicity [32], [33]. Deng et al. [34] were among the first to propose a transformer-based end-to-end visual grounding network, namely TransVG. TransVG [34] employs transformers to establish multimodal correspondences and directly pinpoint referent regions by regressing box coordinates. Nevertheless, the primary fusion transformer in TransVG operates independently from the multimodal encoders, which necessitates training on limited visual grounding data, resulting in optimization challenges and suboptimal performance. To tackle this issue, Deng et al. [35] introduced

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TransVG++, an entirely transformer-based model that integrates BERT [36], vision transformer [37], and a multimodal fusion transformer (MFT). Li and Sigal [32] employed transformer encoders to enhance both visual and textual features. They also developed a text encoder and decoder that can be applied to multiple downstream tasks. Du et al. [38] proposed VGTR, a transformer-based network architecture capable of learning visual characteristics guided by linguistic expressions. The latest transformer-based approaches have begun focusing on improving the visual branch and modifying visual characteristics through the integration of multimodal features. VLTVG [39] seeks to enhance visual features by employing a vision-language verification module and consolidates visual context through a context encoder guided by language. QRNet [40] incorporated a dynamic attention mechanism that is aware of language queries, along with a language query-aware multiscale fusion, for the purpose of adapting visual features.

B. Visual Grounding in RS Images

Unlike images in natural scenes, RS images are characterized by numerous targets and cluttered backgrounds, making specific objects mentioned in queries less prominent. In addition, RS images often have low interclass similarity among targets, presenting highly similar objects across various dimensions, posing significant challenges to the RSVG task. Although natural scene visual grounding has been extensively studied, RSVG remains underexplored. Sun et al. [14] initially introduced the concept of RSVG and introduced an innovative approach, GeoVG. This approach comprises three essential modules: a linguistic encoder, an image encoder, and a fusion module. The linguistic encoder is employed for acquiring numerical geospatial relations and representing intricate expressions as geospatial relation graphs. The image encoder integrates an adaptive regional attention module to discern crucial visual content in large-scale scenes. The fusion module combines text and image features to achieve the ultimate localization. Zhan et al. [13] introduced a new transformer-based RSVG framework, MGVLF. Addressing the scale variation in RS images, MGVLF leverages visual features at multiple scales and text embeddings at various granularities to generate more discriminative representations. To tackle the issue of cluttered backgrounds and noise in RS images, MGVLF dynamically eliminates extraneous noise while amplifying significant features. However, both methods overlook the possibility that a vision backbone lacking query information guidance may fail to effectively focus on specific targets mentioned in queries, and they lack interaction between multimodal data, leading to reduced model performance.

III. Methods

In this section, we will provide a detailed introduction to QAMFN for RSVG. QAMFN is an end-to-end RSVG network consisting of three main components: the linguistic branch, the visual branch, and the multimodal decoder. First, in Section III-A, we will outline the overall architecture of QAMFN. Following this, in Section III-B, we will elaborate on the design of the QGVA mechanism. Section III-C will delve into the design of the text-semantic attention-guided masking (TAM). In Section III-D, we explore the architecture of the MFT and the prediction module. Finally, Section III-E will detail the loss functions used in the training framework.

A. Overview

RSVG is the task of locating target objects in RS images described by text queries. Existing RSVG methods use independent text and image encoders to extract textual and visual features, which are generally global representations. These representations fail to capture local, fine-grained details, leading to a loss of early-stage complementary information between text and image, which in turn limits the model's ability to capture detailed information. Additionally, RS images often contain a large amount of redundant information that is irrelevant to the query text. Current methods lack effective means to suppress the expression of this irrelevant information, resulting in suboptimal performance in complex RS scenarios. To address these issues, we propose a new RSVG framework called QAMFN, as shown in Fig. 3. In this framework, the visual encoding branch starts to incorporate guidance from the query text early on, maximizing the use of complementary information between text and images. During this process, it fully leverages the positional attributes, relational features, and other deep semantic information contained within the text features, ensuring that the visual encoder focuses as much as possible on areas relevant to the query text. Additionally, QAMFN integrates textual and visual features to generate attention masks, effectively suppressing the expression of background information and interfering targets in images.

As shown in Fig. 3, the QAMFN consists of three main parts consisting of a linguistic branch, a visual branch, and a multimodal decoder.

1) Linguistic Branch: The linguistic branch encodes the query text into textual features and is composed of the input embedding and the linguistic backbone. We opted for the pretrained uncased base BERT [36] for the linguistic backbone, which consists of 12 transformer layers [41]. Regarding the expressions, we start by embedding each word into a one-hot vector. Afterward, we convert each one-hot vector into a language token and include the [CLS] and [SEP] tokens. Finally, we extract the language features $F_l \in \mathbb{R}^{l\times d}$ using a BERT-based language backbone, where l and d correspond to the count of language feature tokens and their respective dimensions [42].

2) Visual Branch: The visual branch encodes RS images into query-guided visual features, consisting of a visual backbone and the QGVA module embedded within it. We utilized ResNet [43] as our visual backbone, with the exclusion of the final average pooling and fully connected layers. We divide ResNet into four stages, and at the end of each stage, we introduce a QGVA module to facilitate the integration of visual and linguistic information. The QGVA module takes the visual features extracted from the previous stage and text features as inputs, generates query-guided visual features, and then inputs these into the next stage of ResNet. The input of the visual

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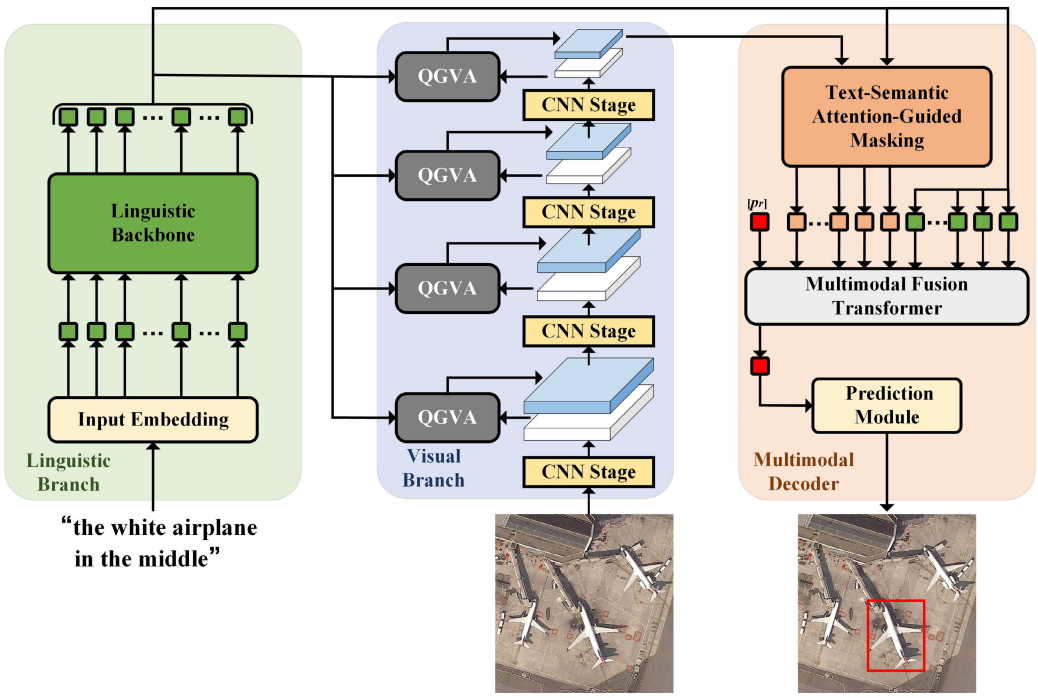


Fig. 3. **Overall architecture of our proposed QAMFN.** The QAMFN consists of three main components: linguistic branch, visual branch, and multimodal decoder. Linguistic branch consists of input embedding and linguistic backbone that transforms the query text into textual features. Visual branch consists of a ResNet and four QGVA modules. QGVA is applied after each stage of the ResNet, utilizing textual features to adjust the visual feature maps. Multimodal decoder consists of **text-semantic attention-guided masking (TAM)**, MFT, and prediction module. The encoded text features and visual features are further fused and decoded to predict the final result. The small green squares represent encoded text tokens, the beige squares represent encoded visual tokens, and the red squares represent learnable tokens used for prediction.

branch is an RS image $I \in \mathbb{R}^{H\times W\times 3}$ and the output is a low-resolution feature map $F_v \in \mathbb{R}^{(H/32)\times(W/32)\times C}$ enriched with query information, where H and W denote the height and width of the input RS image and C represents the channel count of the output feature map.

3) Multimodal Decoder: The multimodal decoder consists of the text-semantic attention-guided masking module, the MFT, and the prediction module. It further integrates and aligns visual and textual features, utilizing the fused features to infer the final results. The input to the multimodal decoder consists of query-guided visual features F_v and textual features F_l, which are first fed into the TAM module. TAM effectively utilizes textual features to collate the contextual visual information regarding target objects. It generates attention masks that are applied to visual feature maps, aiming to diminish the interference caused by cluttered backgrounds and distracting targets in the localization process.

Following this, the visual features enhanced by the TAM and the textual features undergo dimension alignment through two linear projections. After projection, the visual tokens and textual tokens are represented as $f_v \in \mathbb{R}^{N_v\times l}$ and $f_l \in \mathbb{R}^{N_l\times l}$, respectively. Subsequently, a learnable token is appended and concatenated with f_v and f_l. The resulting sequence is as follows:

$$f = \left[\underbrace{f_v^1, f_v^2, \ldots, f_v^{N_v}}_{\text{visual tokens } f_v}, \underbrace{f_l^1, f_l^2, \ldots, f_l^{N_l}}_{\text{textual tokens } f_l}, f_t\right] \tag{1}$$

where $f_t \in \mathbb{R}^{1\times l}$ is the learnable token, which undergoes random initialization prior to the training process. Then, $f \in \mathbb{R}^{(N_v+N_l+1)\times l}$ is encoded into the final fused features by the MFT, which consists of six transformer encoder layers. Finally, the fused features are fed into the prediction module, which is a multilayer perceptron, to predict the bounding box coordinates.

Subsequently, we will provide detailed descriptions of the QGVA module and the TAM module.

B. Query-Guided Visual Attention Module

In this section, we will provide a detailed introduction to the QGVA module. Previous RSVG methods typically involve interactions between the global visual features output by the visual backbone and textual features. However, some key semantic information contained in the textual features may be difficult to reflect in the global visual features, and such crucial information might be filtered out during the process of visual feature extraction. This issue leads to poor alignment of subsequent multimodal features, thereby limiting the model's performance. To address this problem, we designed the QGVA module with the intention of aligning visual features with textual features at an early stage, thereby maximizing the retention of complementary information between visual and textual features. The QGVA module is designed to incorporate linguistic guidance during the image feature extraction process, allowing early visual features to gradually align with textual features to optimally preserve the complementary information between text and images. The QGVA module

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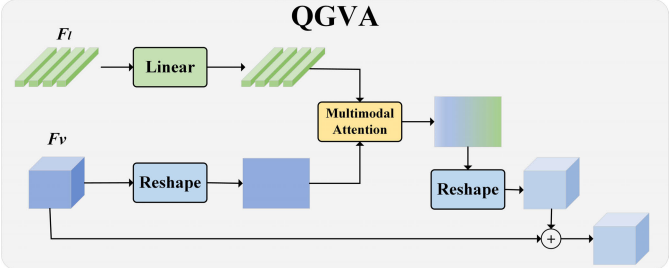


Fig. 4. **Illustration of our QGVA module**, designed to enhance information exchange between visual and linguistic domains by producing visual feature maps that are refined based on textual characteristics.

is placed after each stage of the visual backbone network, taking as inputs the textual features and the visual feature map generated by the previous stage, and outputting a query-guided visual feature map, which then serves as the input for the next stage. Next, we will detail the internal structure of the QGVA module.

As illustrated in Fig. 4, QGVA first maps both linguistic and visual features to the same dimensional space. It then employs multimodal attention to model the process of adjusting image features based on textual features. The multimodal attention consists of a multihead cross-attention (MHCA) module. Within this MHCA module, attention is calculated h times, representing the number of heads. Each individual attention computation involves the inputs: query Q, key K, and value V. The calculation for a single attention instance is performed as follows:

$$\text{Att}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2}$$

where $Q \in \mathbb{R}^{N_Q \times d}$, $K \in \mathbb{R}^{N_K \times d}$, and $V \in \mathbb{R}^{N_K \times d}$. N_Q is the length of Q, N_K is the length of K, and V. d_K is the dimension of K. The output of $\text{Att}(\cdot)$ is the same size as Q. The multimodal attention mechanism is capable of simultaneously focusing on different positions within an input sequence, thereby capturing a richer context. This is particularly beneficial in processing and interpreting complex relationships when dealing with linguistic and visual features. Such a mechanism enhances the understanding and representation of intricate interdependencies inherent in multimodal data.

In the following, we detail the specifics of the QGVA using the ith stage of the CNN as an example. Initially, we map $F_{vi} \in \mathbb{R}^{C_v \times H \times W}$ and $F_{li} \in \mathbb{R}^{N \times L}$ to a common space with C_v dimensions, where $F_{vit} = \theta(\text{flatten}(F_{vi}))$ and $F_{lit} = \theta(\text{linear}(F_{li}))$. θ denotes the tanh activation function, with flatten and linear representing the flattening operation and a linear function, respectively. H and W represent the height and width of image features, while N and L denote the number of tokens and their dimensions. Subsequently, we employ $F_{vit} \in \mathbb{R}^{(H \times W) \times C_v}$ as the query and $F_{lit} \in \mathbb{R}^{N \times C_v}$ as both key and value, inputting them into the multimodal attention module. The process leads to the formation of the fused multimodal feature $F_{vli} \in \mathbb{R}^{(H \times W) \times C_v}$. The mathematical representation of this process is as follows:

$$F_{vit} = \theta(\text{flatten}(F_{vi})), \quad F_{lit} = \theta(\text{linear}(F_{li})) \tag{3}$$

$$F_{vil} = \mathcal{F}_{\text{MHSA}}(Q = F_{vit}, K = F_{lit}, V = F_{lit}) \tag{4}$$

where $\mathcal{F}_{\text{MHSA}}$ represents MHCA. Finally, we summed the original visual features F_{vi} and F_{vli} to get the final visual feature $\tilde{F}_{vli} \in \mathbb{R}^{C_v \times H \times W}$

$$\tilde{F}_{vli} = F_{vli} + F_{vi}. \tag{5}$$

As illustrated in Fig. 3, the QGVA is executed at the conclusion of each stage within the CNN.

C. Text-Semantic Attention-Guided Masking Module

In the RSVG task, the cluttered backgrounds and numerous distractor targets in RS images can impact the reasoning and prediction process. Researchers often design multimodal fusion modules to align visual features with textual features to mitigate the interference caused by redundant information in the visual features. Current RSVG methods typically use stacked attention modules to fuse deep visual and textual features. However, this fusion method lacks flexibility and fails to fully exploit the deep semantic associations between visual and textual features, limiting the model's multimodal reasoning capabilities. To address this issue, we propose the TAM module. TAM uses a dynamic fusion method to align and integrate multimodal features. By dynamically adjusting the visual features based on the textual features, TAM aligns multimodal features while retaining only the parts of the visual features relevant to the textual features, thus achieving multimodal feature fusion and filtering out redundant features.

As shown in Fig. 5, the TAM consists of four stages: first, feature preprocessing and projection, where image and text features are unified into the same dimensional space. Second, independent modality deep feature extraction, using Transformer encoders to thoroughly analyze and extract detailed features of each modality. Third, interactive attention fusion, which employs attention mechanisms to emphasize regions in the image matching the textual description. Finally, dynamic feature adjustment and final fusion, where the text influenced dynamic image adapter (TIDIA) dynamically adjusts image features based on the textual content, and a Transformer decoder integrates these features to generate the attention mask. Afterward, we will introduce several important submodules in the TAM module, followed by a detailed explanation of the TAM module's framework.

1) Self-Attention Layer: Self-attention (SA) layer contains a multihead SA (MHSA) module, a feed-forward network (FFN), and two layer normalization (LN) [44] blocks with residual connection. The MHSA and MHCA share an identical structure, with the key distinction being that MHCA utilizes different sequences for Q, K, and V, while MHSA employs the same sequence for all three. Assuming the input sequence is $\mathbf{x}_n$, the process of SA can be described as follows:

$$\boldsymbol{x}'_n = \text{LN}(\boldsymbol{x}_n + \mathcal{F}_{\text{MHSA}}(\boldsymbol{x}_n)) \tag{6}$$

$$\boldsymbol{x}_{n+1} = \text{LN}\left(\boldsymbol{x}'_n + \mathcal{F}_{\text{FFN}}(\boldsymbol{x}'_n)\right). \tag{7}$$

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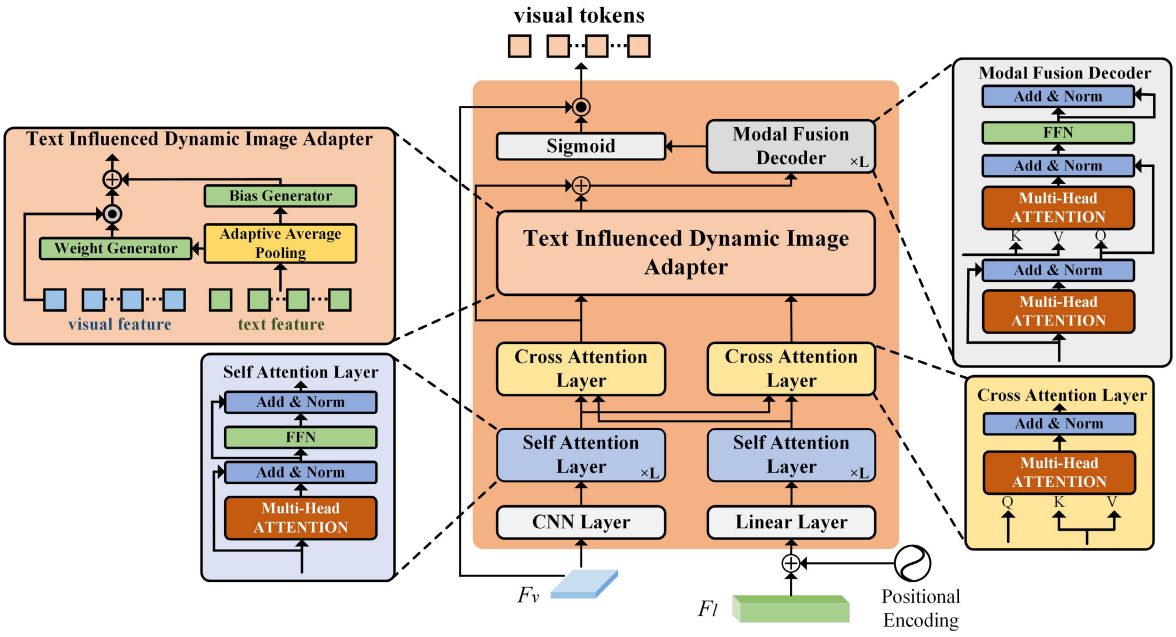


Fig. 5. **Illustration of our proposed text-semantic attention-guided masking module.** Text-semantic attention-guided masking module comprises four main submodules: SA layer, CA layer, TIDIA, and MFD.

2) Cross-Attention Layer: Cross-attention (CA) layer contains an MHCA module and an LN blocks with residual connection. For the query tokens $\boldsymbol{x}_n$ and support tokens $\boldsymbol{y}_n$, the process of cross-modal attention can be described as follows:

$$\boldsymbol{x}_{n+1} = \mathrm{LN}\big(\boldsymbol{x}_n + \mathcal{F}_{\mathrm{MHCA}}(\boldsymbol{x}_n, \boldsymbol{y}_n, \boldsymbol{y}_n)\big). \tag{8}$$

3) Text Influenced Dynamic Image Adapter: To ensure the quality of the fused features and further refine them, we propose the TIDIA. This adapter constructs a linear layer that generates parameters using textual features to process visual features. It dynamically adjusts the visual features based on a global representation of the textual features. Such adjustments allow for a more precise reflection of the correlation between text and images. The TIDIA consists of an adaptive average pooling layer, a weight generator, and a bias generator, where both the weight and bias generators are linear layers. The global average representation of textual features is obtained through the adaptive pooling layer, followed by the generation of weights and biases using the weight generator and bias generator, respectively. These weights and biases serve as parameters that are applied to the image features, resulting in refined fused features. Assuming the input image features are denoted as $\boldsymbol{x}_n$ and the textual features as $\boldsymbol{y}_n$, the process of the TIDIA can be described as follows:

$$\boldsymbol{y}_{\mathbf{avg}} = \mathrm{AdaptiveAvgPool}(\boldsymbol{y}_n) \tag{9}$$

$$\boldsymbol{W} = \mathrm{Weight_Generator}\big(\boldsymbol{y}_{\mathbf{avg}}\big) \tag{10}$$

$$\boldsymbol{B} = \mathrm{Bias_Generator}\big(\boldsymbol{y}_{\mathbf{avg}}\big) \tag{11}$$

$$\boldsymbol{x}_{\boldsymbol{n+1}} = \boldsymbol{W}\boldsymbol{x}_{\boldsymbol{n}} + \boldsymbol{B} \tag{12}$$

where $\boldsymbol{W}$ and $\boldsymbol{B}$ are the weights and biases generated from the textual features.

4) Modal Fusion Decoder: We further integrate the adjusted fused features using the modal fusion decoder (MFD). MFD contains an MHSA module, an MHCA module, two FFNs, and three LN blocks with residual connection. For the input sequences $\boldsymbol{x}_n$ and $\boldsymbol{y}_n$, the process of the MFD can be represented as follows:

$$\boldsymbol{x}'_n = \mathrm{LN}(\boldsymbol{x}_n + \mathcal{F}_{\mathrm{MHSA}}(\boldsymbol{x}_n)) \tag{13}$$

$$\boldsymbol{x}''_n = \mathrm{LN}\big(\boldsymbol{x}'_n + \mathcal{F}_{\mathrm{MHCA}}(\boldsymbol{x}'_n, \boldsymbol{y}_n, \boldsymbol{y}_n)\big) \tag{14}$$

$$\boldsymbol{x}_{n+1} = \mathrm{LN}\big(\boldsymbol{x}''_n + \mathcal{F}_{\mathrm{FFN}}(\boldsymbol{x}''_n)\big). \tag{15}$$

Given the image features $F_v \in \mathbb{R}^{C\times H\times W}$ and the text features $F_l \in \mathbb{R}^{N\times L}$ as inputs, where C represents the number of channels in the feature map, H and W represent the height and width of the feature map, respectively, N represents the number of tokens in the text features, and L represents the length of the tokens. First, we process F_v with a 1×1 convolution and a flatten operation to obtain $F'_v \in \mathbb{R}^{C\times(H\times W)}$, enabling it to adapt to the input of the subsequent transformer structure, and we use a linear function to process F_l, resulting in $F'_l \in \mathbb{R}^{N\times L_v}$. The length of the tokens L_v in F'_l has the same dimension as $(H \times W)$. Second, we use two independent SA modules to process F'_v and F'_l separately, resulting in F_{vd} and F_{ld}. Each SA module consists of two layers of SA layers. As shown in Fig. 5(c), we use two CA layers to enhance the interaction between multimodal data. The left CA layer takes F_{vd} as the query and F_{ld} as the key and value, resulting in text-guided image features F_{vl}, which have the same dimensions as F_{vd}. The right CA layer takes F_{ld} as the query and F_{vd} as the key and value, resulting in image-guided text features F_{lv}, which have the same dimensions as F_{ld}. Then, F_{lv} generates weights W and bias B through the weight generator and the bias generator. The refined fusion feature F'_{vl} is obtained by element-wise multiplication of F_{vl} with the weight W, followed by adding the bias B. F'_{vl} and F_{vl} have the same dimensions. Finally, we use the MFD to further integrate

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the fusion features. F'_{vl} is input into the MFD, and F_{vl} is used as the key and value for the second MHCA module, resulting in the final fusion feature $F_{vlf} \in \mathbb{R}^{C_m \times L_v}$. After being reshaped, F_{vlf} is processed through a sigmoid function, resulting in the final attention mask $F_{\text{mask}} \in \mathbb{R}^{C \times H \times W}$. F_{mask} is multiplied element-wise with the original image features F_v to obtain the final visual features.

IV. Experiments

In this section, we conduct comprehensive experiments to demonstrate the benefits of our proposed QAMFN. Sections IV-A and IV-B introduce the experimental setup, datasets, and evaluation metrics in detail. Section IV-C presents the experimental results of our QAMFN and compares it with other SOTA methods. In Section IV-D, we conduct comprehensive ablation experiments to validate the efficacy of our QAMFN. Finally, Section IV-E showcases some qualitative results to provide a comprehensive analysis of our model.

A. Datasets and Evaluation Metric

1) Datasets: We selected the widely used DIOR-RSVG [13] dataset in the field of RSVG to train and test our model. The DIOR-RSVG [13] builds upon the original DIOR dataset [48]. It encompasses 38 320 language expressions linked to 17 402 RS images, each measuring 800 × 800 pixels. The expressions average 7.47 words in length, drawn from a vocabulary of 100 words. This dataset covers 20 different object categories. For our experimental setup, we divided the dataset into training, validation, and test sets, allocating 40%, 10%, and 50% of the expressions and their associated images to each set, respectively, using a random assignment approach.

2) Evaluation Metric: For a given RS image and query, we consider the prediction to be correct when the IoU of the prediction box with the ground-truth bounding box exceeds a certain threshold. Our evaluation metrics align with those used in MGVLF [13], encompassing precision at IoU thresholds of 0.5, 0.6, 0.7, 0.8, and 0.9 (Pr@0.5, Pr@0.6, Pr@0.7, Pr@0.8, and Pr@0.9). Additionally, to effectively evaluate the alignment between the predicted bounding boxes and the ground truth, we have also incorporated the mean IoU and cumIoU [49] as evaluation metrics. The formulas for these metrics are as follows:

$$\text{meanIoU} = \frac{1}{N} \sum_n I_n / U_n \tag{16}$$

and

$$\text{cumIoU} = \left(\sum_n I_n \right) \Big/ \left(\sum_n U_n \right) \tag{17}$$

where n denotes the index of an individual sample, and N represents the total number of samples in the dataset. I_n and U_n are the areas of intersection and union between the predicted box and the ground-truth box, respectively.

B. Implementation Details

The image dimensions are standardized to a fixed size of 640 × 640 for training. The maximum length of language tokens is constrained to Nt = 40, with a dimension of D = 256. A pretrained ResNet-50 serves as the visual backbone, while the MFT module utilizes the pretraining weights of the DETR model [50]. For the linguistic aspect, pretraining weights from the BERT model [36] are employed. BERT's hidden size b stands at 768. The MGVLF [13] framework is followed for processing the input images and expressions. AdamW [51] with a weight decay of $1e^{-4}$ is selected as the optimizer during training. The network is trained over 120 epochs with a batch size of 20 on two NVIDIA GeForce RTX 3090 GPUs, each equipped with 24 GB of VRAM. In the transformer, the dropout ratio for FFN is maintained at 0.1. The initial learning rate for pretrained parameters in the network is set at $1e^{-5}$ and $1e^{-4}$ for other parameters. Xavier [52] is utilized to randomly initialize the parameters in the network that lack pretraining.

Continuing from the previous method [34], we apply the commonly used smooth L1 loss [53] $\mathcal{L}_{\text{smooth}-L1}(\cdot)$ and the generalized intersection over union (GIoU) loss [54] $\mathcal{L}_{\text{GIoU}}(\cdot)$ on the 4-D bounding box coordinate. The comprehensive loss function for training our proposed network can be formulated as follows:

$$\mathcal{L} = \mathcal{L}_{\text{smooth}-L_1}(\boldsymbol{b}, \hat{\boldsymbol{b}}) + \mathcal{L}_{\text{GIoU}}(\boldsymbol{b}, \hat{\boldsymbol{b}}) \tag{18}$$

where $\boldsymbol{b}$ indicates the coordinates of the correct bounding box and $\hat{\boldsymbol{b}}$ denotes the coordinates of the bounding box predicted by the model.

C. Comparisons With SOTA Methods

The comparative results of our proposed QAMFN with conventional natural image visual grounding methods and RSVG approaches are presented in Table I. Experimental outcomes demonstrate that QAMFN outperforms other methods in all metrics except for Pr@0.9, achieving 81.67% in Pr@0.5, which is a 4.98% enhancement compared to the current SOTA RSVG approache MGVLF [13]. The top part of Table I showcases the performance of two-stage methods in RSVG tasks. These two-stage methods rely on pretrained object detectors to generate proposal boxes, a process that does not take into account the rich information provided by textual queries, resulting in an inability to efficiently generate effective candidate regions, thus limiting their performance in RSVG tasks. The middle section of Table I displays the outcomes of current one-stage methods. These approaches require the development of complex, manually designed visual–language fusion modules to produce bounding boxes, which can result in underutilization of multimodal information and overfitting to specific datasets. In contrast, our approach utilizes a transformer-based framework, allowing for a simpler yet more comprehensive integration of visual and language components. Except for Pr@0.9, our method significantly surpasses one-stage approaches in other metrics. FAOA [27] achieves the best performance at Pr@0.9 by fusing

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TABLE I

COMPARISON WITH STATE-OF-THE-ART METHODS. THE BEST PERFORMANCE IS IN BOLD, AND THE SECOND PERFORMANCE IS UNDERLINED. SOME RESULTS ARE FROM [13]

Methods	Venue	Visual Encoder	Language Encoder	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	meanIoU	cumIoU
Two-stage										
MattNet [20]	*CVPR'18*	ResNet-101	BiLSTM	70.16	63.79	56.82	45.04	16.45	60.32	70.13
NMTree [22]	*ICCV'19*	ResNet-101	BiLSTM	68.12	63.09	56.71	40.04	11.64	59.28	69.11
One-stage										
ZSGNet [45]	*ICCV'19*	VGG	BiLSTM	48.12	43.79	36.82	25.04	6.64	40.23	46.11
ZSGNet [45]	*ICCV'19*	ResNet-50	BiLSTM	51.67	48.13	42.30	32.41	10.15	44.12	51.65
FAOA [27]	*ICCV'19*	DarkNet-53	BERT	67.21	64.18	59.23	50.87	34.44	59.76	63.14
FAOA [27]	*ICCV'19*	DarkNet-53	LSTM	70.86	67.37	62.04	53.19	**36.44**	62.86	67.28
ReSC [30]	*ECCV'20*	DarkNet-53	BERT	72.71	68.92	63.01	53.70	33.37	64.24	68.10
LBYL-Net [46]	*CVPR'21*	DarkNet-53	LSTM	73.29	69.92	63.97	48.07	16.60	65.86	75.45
LBYL-Net [46]	*CVPR'21*	DarkNet-53	BERT	73.78	69.22	65.56	47.89	15.69	65.92	76.37
Transformer-based										
VGTR [38]	*ICME'22*	ResNet-50	LSTM	68.12	62.73	53.46	41.52	20.54	56.96	67.46
SeqTR [47]	*ECCV'22*	DarkNet-53	LSTM	71.29	65.83	58.35	46.47	23.78	61.76	72.33
TransVG [34]	*ICCV'21*	ResNet-50	BERT	72.41	67.38	60.05	49.10	27.84	63.56	76.27
VLTVG [39]	*CVPR'22*	ResNet-50	BERT	69.41	65.16	58.44	46.56	24.37	59.96	71.97
VLTVG [39]	*CVPR'22*	ResNet-101	BERT	75.79	72.22	66.33	55.17	33.11	66.32	77.85
MGVLF [13]	*TGRS'23*	ResNet-50	BERT	76.78	72.68	66.74	56.42	35.07	68.04	78.41
QAMFN(Ours)	-	ResNet-50	BERT	**81.67**	**75.15**	**68.56**	**57.43**	34.51	**71.78**	**84.55**

TABLE II

ABLATION EXPERIMENT OF THE MAIN MODULES

Cases	QGVA	TAM	MFT	Pr@0.5(%)
(a)	-	-	✓	72.41
(b)	✓	-	✓	78.29(+5.88)
(c)	-	✓	✓	78.19(+5.78)
(d)	✓	✓	-	77.93(+5.52)
Ours	✓	✓	✓	**81.67(+9.26)**

TABLE III

COMPARATIVE ABLATION STUDY WITH MGVLF

Cases	Base Model	QGVA	Pr@0.5(%)
(a)	MGVLF[13]	-	76.78
(b)	QAMFN	-	78.19(+1.41)
(c)	MGVLF[13]	✓	79.85(+3.07)
(d)	QAMFN	✓	**81.67(+4.89)**

textual embeddings into YOLOv3 and integrating multimodal features at different spatial resolutions.

Our method was also compared with other transformer-based approaches. VGTR [38] is an end-to-end visual grounding framework based on transformers. It uses transformer encoders and decoders to fuse multimodal data and predict the final result. The multimodal data fusion part in VGTR is relatively simple and crude, which prevents it from achieving optimal performance. SeqTR [47] transforms visual grounding into a point prediction problem with image and text inputs as conditions, representing bounding boxes as a series of discrete coordinate markers. Breaking down bounding boxes into discrete coordinates may lead to a decrease in accuracy, as prediction errors can accumulate and result in inaccurate localization. TransVG [34] designs an effective visual grounding architecture by stacking transformers. However, its core fusion transformer operates independently of unimodal encoders, limiting its performance. VLTVG [39] introduces a language-guided visual feature aggregation method and a multilevel cross-modal decoder for accurate visual grounding. Yet, its effectiveness is constrained by the quality of visual features extracted by the visual backbone. When the backbone is ResNet-50, VLTVG [39] performs less effectively than TransVG [34]. MGVLF [13] operates on the ultimate output of the visual backbone across various scales, enhancing its ability to acquire discriminative visual representations for fusion with textual modalities. However, its insufficient utilization of textual information limits its ability to effectively distinguish between distracting targets in RS images, hence constraining its localization performance. Our QAMFN leverages a QGVA mechanism to actively direct the visual backbone toward query-relevant visual information. Additionally, we designed a textual-semantic attention-guided masking module (TAM) that enhances the model's ability to distinguish high-similarity targets by generating attention masks to suppress background information and distractors.

D. Ablation Study

In this section, we carried out a series of experiments on Dior-RSVG to confirm the efficacy of our proposed QAMFN.

1) Ablation Experiments for the Entire Network: Table II lists the results of our ablation studies on the three main components of QAMFN: the QGVA module, the text-semantic attention-guided masking (TAM) module, and the MFT. In Case (a), QAMFN degrades to TransVG [34] without QGVA and TAM, achieving only 72.41% accuracy on the test set. In Case (b), we introduced QGVA, enabling the visual backbone to actively perceive target objects based on query information, and generating visual features that are more conducive to subsequent multimodal inference. The accuracy improved by 5.88% compared to Case (a). In Case

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TABLE IV
ABLATION EXPERIMENT ON THE NUMBER OF MULTIHEAD ATTENTION LAYERS IN THE QGVA

Cases	N_m	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	meanIoU	cumIoU
(a)	3	74.72	69.41	61.81	48.96	26.04	65.11	78.14
(b)	2	79.26	72.12	62.18	55.48	27.94	69.46	80.78
Ours	1	**81.67**	**75.15**	**68.56**	**57.43**	**34.51**	**71.78**	**84.55**

TABLE V
ABLATION EXPERIMENT ON THE NUMBER OF TIDIA LAYERS IN THE TAM

Cases	C_m	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	meanIoU	cumIoU
(a)	0	76.82	68.37	56.78	44.36	19.43	59.11	69.50
(b)	1	79.61	72.20	63.81	52.30	22.79	67.60	76.58
(c)	2	80.23	73.42	65.33	54.18	25.49	69.60	79.81
Ours	3	**81.67**	**75.15**	**68.56**	**57.43**	**34.51**	**71.78**	**84.55**

TABLE VI
ABLATION STUDIES OF THE QGVA MODULE

Cases	Stage1	Stage2	Stage3	Stage4	Pr@0.5 (%)
(a)	-	-	-	-	78.19
(b)	-	-	-	✓	78.93(+0.74)
(c)	-	-	✓	✓	79.21(+1.02)
(d)	-	✓	✓	✓	80.64(+2.45)
Ours	✓	✓	✓	✓	**81.67(+3.48)**

(c), we introduced the TAM module, enhancing the overall multimodal inference capability of the model, resulting in a 5.78% accuracy improvement compared to Case (a). In Case (d), we introduced both QGVA and TAM while removing the original MFT module, achieving an accuracy of 77.93%, which is a 5.52% improvement over Case (a). The last row of Table II shows the results of the complete QAMFN, achieving an accuracy of 81.67%, representing the best performance. The visualized feature map is shown in Fig. 6. The comparison of Fig. 6(a) and (b), as well as the comparison of Fig. 6(c) and (e), both illustrate that QGVA can indeed help the visual backbone to customize visual features that are more relevant to the query information. The comparison of Fig. 6(a) and (c) can also illustrate that TAM does have the ability to suppress the expression of background information and interfering targets in visual features.

2) Comparative Ablation Study With MGVLF: To further validate the effectiveness of our proposed framework, we introduced the QGVA mechanism into MGVLF, the SOTA method previously in the RSVG domain, and conducted experiments. The experimental results are presented in Table III. The comparison between Case (a) and Case (c) demonstrates that incorporating the QGVA mechanism into MGVLF leads to a significant performance improvement, validating the effectiveness of our proposed framework. Introducing query information guidance into the visual backbone is beneficial for enhancing the model's performance. Additionally, the comparisons between Case (a) and Case (b), as well as between Case (c) and Case (d), also highlight that the performance of our proposed TAM module surpasses that of the cross-modal interaction component in MGVLF, further confirming the robust capabilities of the TAM module.

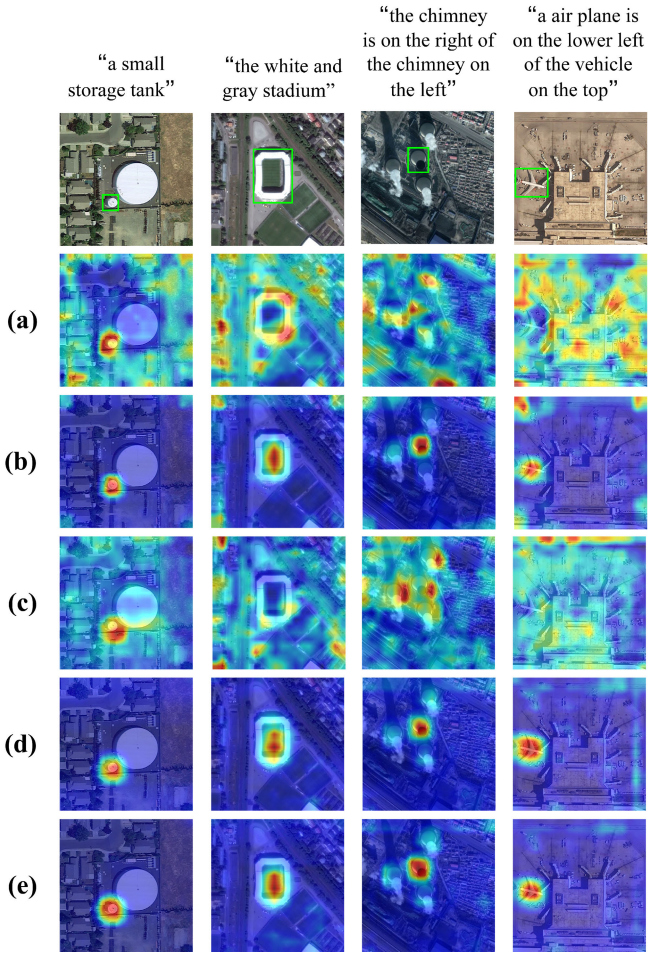
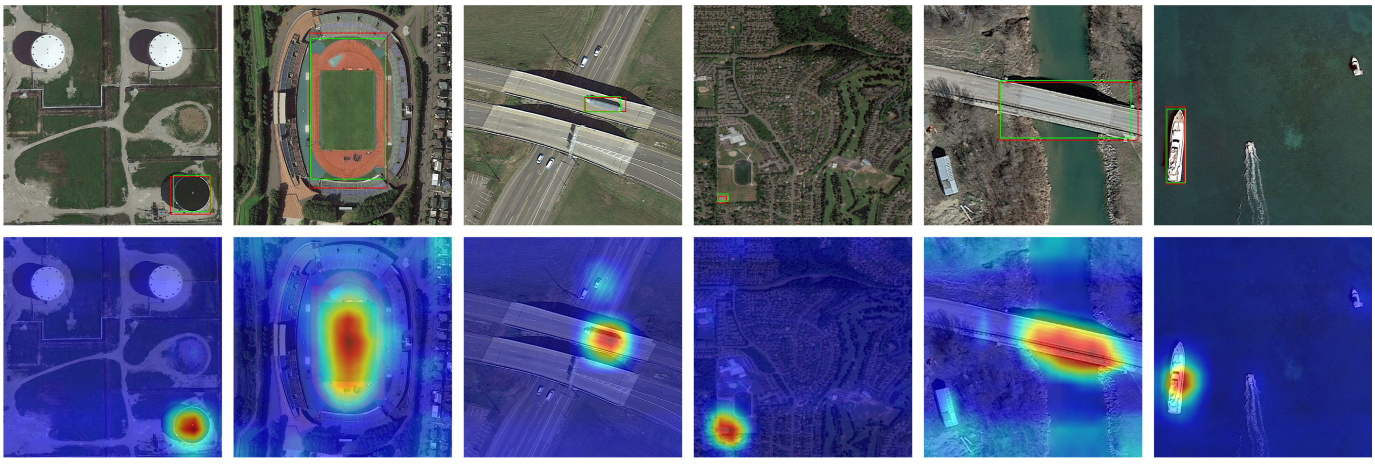


Fig. 6. Visualization of the ablation experiment for the main modules of QAMFN. In the top two lines of the image, the text within the black box in the first line represents a query for the image below, where the target referred to by the query is outlined by a green box. (a)–(d) Attention maps for the (a)–(d) cases in Table II, while (e) represents the attention map for our approach.

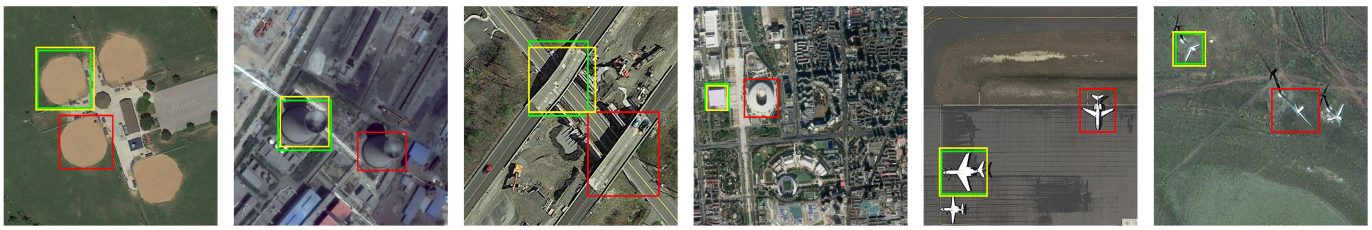
3) Ablation Experiments on QGVA: To further validate the effectiveness of the QGVA module, we positioned it

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(a) (b) (c) (d) (e) (f)

Fig. 7. Visualization of the final grounding results and the attention maps of our proposed QAMFN. In the first row of the picture, the green boxes represent the ground truth and the red boxes are the inference results of QAMFN. The second row shows the corresponding feature maps. (a) "Gray and black storage tank." (b) "Ground track field is in the gray stadium." (c) "Large vehicle." (d) "Baseball field is on the lower left of the baseball field on the left." (e) "Slender gray large bridge." (f) "Large ship."



(a) (b) (c) (d) (e) (f)

Fig. 8. Some grounding results for images containing high-similarity distractor objects are presented. In the images, the green boxes represent the ground-truth bounding boxes, the yellow boxes denote the inference results of our method, and the red boxes illustrate the inference results of the previous RSVG method. (a) "Baseball field in the middle." (b) "Large chimney." (c) "Overpass in the middle." (d) "Stadium on the left." (e) "Large airplane." (f) "Windmill on the upper left."

after different stages of ResNet and observed the outcomes, as illustrated in Table VI. In Case (a), we removed the QGVA module from the network, resulting in a Pr@0.5 of only 78.19%. In Case (b), we added a QGVA module after the last stage of ResNet, which increased the Pr@0.5 by 0.74% compared to Case (a). In Case (c), we added QGVA modules to the last two stages of ResNet, achieving a Pr@0.5 of 79.21%. In Case (d), we added QGVA modules to the last three stages of ResNet, resulting in a Pr@0.5 increase of 2.45% compared to Case (a). When QGVA modules were added to every stage of ResNet, Pr@0.5 reached 81.67%, achieving the best results. These findings indicate that the QGVA module at each stage contributes to the performance improvement of the model, further validating the effectiveness of our proposed framework.

Furthermore, to assess the rational design of the QGVA module, we experimented with varying the number of layers in the multihead attention component. The outcomes are presented in Table IV, where N_m denotes the number of layers in the multihead attention. The model achieved the best performance across all metrics when N_m was set to 1. A decrease in all metrics was observed when N_m was increased to 2, and a significant decline occurred when N_m was raised to 3. These results suggest that the dataset may not train an excessive number of multihead attention modules effectively. Not only does an abundance of these modules increase experimental costs, but it also diminishes the experimental outcomes.

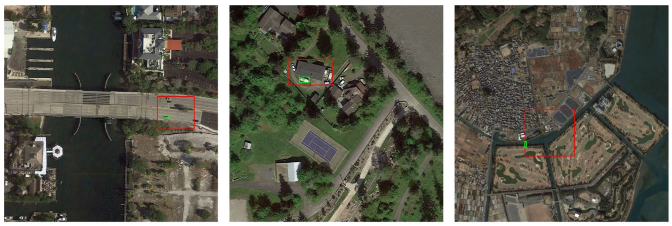
4) Ablation Experiments on TAM: To validate the design of the TAM module, we conducted ablation experiments on each submodule within the TAM module. The experimental results are presented in Table VII. Here, SA represents the self-attention module, CA denotes the cross-attention module, TIDIA stands for text influenced dynamic image adapter, and MFD signifies the modal fusion decoder. In Case (d), where all submodules were included, the experimental outcome was optimal, achieving 81.67% in Pr@0.5. In Case (a), the removal of the SA module led to a decrease of 0.48% in Pr@0.5. In Case (b), eliminating the CA module resulted in a 1.04% reduction in Pr@0.5. In Case (c), the absence of the TIDIA module caused a decline of 4.85% in Pr@0.5. Finally, in Case (d), the removal of the MFD module led to a 0.89% decrease in Pr@0.5. These results indicate that the SA, CA, and MFD modules contribute marginally to model performance, while the TIDIA module significantly impacts overall performance.

To further investigate the influence of the TIDIA module on the overall model, we retained the remaining submodules of TAM and increased the number of layers of TIDIA in our experiments. The results are displayed in Table V, where C_m represents the number of layers in the TIDIA module. In Case (d), the best result, with $C_m = 3$, Pr@0.5 reached

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TABLE VII
ABLATION STUDIES OF THE TAM MODULE

Cases	SA	CA	TIDIA	MFD	Pr@0.5 (%)
(a)	-	✓	✓	✓	81.19(−0.48)
(b)	✓	-	✓	✓	80.63(−1.04)
(c)	✓	✓	-	✓	76.82(−4.85)
(d)	✓	✓	✓	-	80.78(−0.89)
Ours	✓	✓	✓	✓	**81.67**



(a) (b) (c)

Fig. 9. Some failed experimental results. (a) "Vehicle is smaller in size to the vehicle on the upper right." (b) "Ship is on the lower right of the vehicle on the upper left." (c) "Bridge is much smaller than the bridge on the left."

81.67 %. In Case (a), where the TIDIA module was omitted, Pr@0.5 only reached 76.82%. In Case (b), with one layer of TIDIA, Pr@0.5 was 79.61%. In Case (c), with two layers of TIDIA, Pr@0.5 reached 80.23%. These results demonstrate that appropriately increasing the number of layers in the TIDIA module can effectively enhance the overall performance of the model. However, further increasing the number of layers in the TIDIA module, beyond $C_m = 3$, did not result in better model fitting, possibly due to the limited data volume of the dataset.

E. Qualitative Results

In this section, we visualize the results within the test set and engage in a qualitative discussion. As illustrated in Fig. 7, we demonstrate the visual grounding results of the QAMFN model facing targets of various types, along with the presentation of visual feature maps. It can be observed from the visual feature maps that QAMFN effectively focuses on the key targets indicated by the queries, while substantially suppressing the representation of background information and distracting targets. In Fig. 8, we compare the capabilities of the previous RSVG method and QAMFN in handling scenarios with highly similar distracting targets. The experimental results reveal that the previous RSVG method has very limited capacity in differentiating objects of high similarity, failing to effectively distinguish these targets based on the semantic information provided by the query texts. By incorporating query information guidance in the process of extracting visual features, QAMFN customizes visual features highly relevant to the query. With a more powerful cross-modal interaction module, QAMFN can fully utilize deeper target features in the query to effectively differentiate between highly similar targets in the image.

Although the QAMFN has achieved commendable experimental results, there are instances where it fails to perform optimally. As shown in Fig. 9, where the green box represents the ground truth and the red box indicates the results inferred by QAMFN, it is evident that the model struggles with accurately locating smaller targets corresponding to longer texts. The model tends to focus on the general area where the true target is located, yet fails to precisely pinpoint its position. These targets are generally smaller in scale, and their features may become obscured by the background during the process of visual feature extraction, leading to the model's inability to accurately determine their locations. We believe this may be due to the insufficient clarity of small targets in the dataset and the lack of small target samples in the training data. In future research, constructing higher resolution datasets with a wider range of target scales to train better RSVG models is an important research direction.

V. CONCLUSION

In this article, addressing the problem that the pretrained visual backbones do not produce visual features consistent with the requirements of RSVG, we introduce a novel RSVG framework named QAMFN, designed to use query information for guiding the visual backbone toward specific targets. This framework comprises two key modules: the QGVA module leverages query information to enhance visual features, facilitating the visual backbone in iteratively acquiring query-relevant visual characteristics. The TAM module aggregates visual and textual features to generate attention masks, thereby suppressing irrelevant information in the visual features. The experimental findings conclusively establish that our proposed approach surpasses the existing methods in both natural image visual grounding and RSVG, highlighting its remarkable effectiveness and superiority. In the future, we will do more meaningful work on the characteristics of RS scenarios and the various problems that still exist in the current RSVG methods.

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